


# ID6003W - AI in Process and Logistic optimization

## MINI PROJECT

## GROUP 3

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# 1. INTRODUCTION

Industrial control loops are fundamental elements of process automation systems. However, loops often exhibit undesirable behaviors such as oscillations, noise amplification, disturbances, high variance, or valve stiction. Identifying these issues manually is time-consuming and subjective.

This project aims to **automate loop diagnostics** using data-driven methods. The provided loop datasets contain PV (Process Variable), SP (Setpoint), and OP (Controller Output) values. Along with these, we have labeled assessments of oscillatory behavior, noise, disturbance, and stiction probability.

# 2. ARCHITECTURE

The system is orchestrated by *run_pipeline.py,* which executes a sequential pipeline:

- Input Analysis: Validated 100 training loops and identified null label distributions.
- Training Pipeline: Leveraged pre-trained models to skip redundant training steps, optimizing runtime.
- Inference: Applied models to unseen Test Data shared for our group  (Loops 12, 77, 101, 131).
- Post-Processing: Categorized predictions and generated dual-layer text inferences.
- Reporting: Produced outputs/**test_results_with_inferences.csv** which has the inferences drawn from the uploaded test loops and *test_results.csv* with full traceability.

**Key Modules**:

*src/run_pipeline.py*: The central script file integrating all sub-modules.

*src/features.py*: The core feature processing engine.

*src/models.py* :  XGBoost/RandomForest models for classification. , Logistic regression for regression

*src/generate_inferences.py*: Logic for creating textlike rule-based insights.

*src/generate_eda.py* : EDA Code for the loop data

*src/generate_training_report.py :* Comprehensive training logs report

*src/inference.py:* Script to create automatic inferences based on model results

# 3. METHODOLOGY

Analysis of the loop description excel label file detected the following columns:

['Loop No', 'Oscillatory?', 'Disturbance', 'Comments', 'Sticky Valve probability ']

Missing values were minimal and handled through controlled imputation.

## 3.1. Feature Engineering

For each loop, the pipeline computes:

### Statistical Features

- error mean absolute (|SP–PV|)
- error standard deviation
- PV standard deviation
- OP variation statistics

### Spectral Features

- spectral entropy
- PSD dominant frequency

### Temporal Features (index-based)

- error zero crossings
- PV drift slope
- autocorrelation energy (ACF)

### Control Interaction Features

- cross-correlation max value

- OP jump count
- stiction flatness metric

These form a rich representation of dynamic loop behavior.

## 3.2. Label Preprocessing

Labels such as: “Looks oscillatory” , “High variance” , “Yes at PV” , “may be” were mapped into structured classes:

| Behavior | Classes |
|---|---|
| Oscillation | No / Maybe / Yes |
| Disturbance | No / Maybe / Yes |
| Stiction | Continuous (0–1 range) |
| Noise | No / Maybe / Yes |

A label audit confirmed:

- **42%** oscillatory potential loops
- **27%** potentially noisy loops
- **30%** with disturbance indicators

## 3.4. ML Models and Training

The project required evaluating several ML algorithms, and different team members experimented with:

- **RandomForestClassifier**

- **XGBClassifier**
- **GradientBoostingClassifier**
- **SVR (Support Vector Regression)** for stiction prediction

Instead of choosing a single fixed model, we implemented a **model selection framework as part of the end-to-end pipeline creation** :

### For every target (Oscillation, Noise, Disturbance, Variance, Stiction):

1. Train all candidate models
2. Evaluate them using
    - 80:20 train–validation split
    - 5-fold cross-validation
3. Compare evaluation metrics:
    - F1 macro-score for classification tasks
    - RMSE for regression (stiction)

4. **Automatically choose the best-performing model**
5. Store the selected models in `trained_models.pkl`

This approach ensures:

- the system adapts to new datasets
- classification tasks use the strongest model dynamically
- stiction prediction selects between **SVR** and **XGBoost** based on performance

## 7. My Contribution: End-to-End Pipeline Creation and Model Integration

My primary contribution to this project was the complete **design, integration, and operationalization of the automated industrial loop analysis pipeline**.

Different members of the group worked on various components such as feature extraction, label preprocessing, ML model trials, EDA visualization, and inference

scripts. However, these existed as independent Jupyter notebooks and standalone Python blocks and artifacts were shared amongst ourselves .

The team sat down on 6 occasions to discuss the approach to be taken and how features will be created . Post that discussion , there were hiccups while implementing the models and the output results . The observations were shared , new strategies were devised and worked on that . For example : for columns like oscillation , we had to switch between Yes/No/Maybe and Yes/No , as it is not right that we assign Maybe to yes .

I was responsible for taking all of these scattered pieces, standardizing them, and creating a **single-click, production-ready pipeline** capable of:

- reading any number of loop CSV files
- preprocessing labels
- extracting features
- training and selecting the best model dynamically
- generating EDA and feature importance plots
- producing inference results on unseen test loo
- validating ML outputs using physical metrics
- generating final Excel/CSV reports and summaries

I built the outer pipeline that performs the full workflow using a single command:

```
python run_pipeline.py
```

This script handles all 10 stages observed in the execution logs:

1. **Input file analysis**

2. **Training pipeline (or model loading)**

3. **EDA and balance plots**
4. **Feature importance extraction**
5. **Label audit and integrity checks**
6. **Held-out test inference (auto-detect test folder)**
7. **Categorization of severity**
8. **Physical validation (entropy, ACF, OP flatness)**
9. **Training reference data summary**
10. **Severity vs label alignment report**

This ensures that once test loop CSV files are placed in the `/test_data` directory, the entire evaluation happens **without any manual intervention**.

Writing the scripts meant knowing and understanding everyone's work and iterating on top of that . All intermediate steps are saved as pickle files and results from each step are saved as csv files in output .Along with verification logs , addition of visualisation steps are performed and generated figs are saved in

Currently , running ``` python run_pipeline.py ``` will be the oneclick format to run the whole pipeline script . This ensures that once test loop CSV files are placed in the /test_data directory, the entire evaluation happens without any manual intervention.

This script automatically:

- detects the presence of trained models
- trains only if models are missing
- loads test data
- generates inference results
- produces Excel and CSV files
- plots EDA and importance graphs

- logs every step with timestamped messages

This made the system reproducible, debuggable, and user-friendly.

# 4. RESULTS

Across all models, the top discriminative features consistently included:

**For Oscillation Detection** ,

*Error_mean_abs , cc_max_val , pv_drift_slope , error_std , pv_diff_std*

**For Noise Detection**

error_std (dominant indicator of noise), error_zero_crossings , pv_std,

Spectral_entropy ,

**For Stiction Probability**

Spectral_entropy , psd_dom_freq ,op_jump_count , cc_max_val , err_acf_energy

These results validate that the engineered features encode both time-domain and frequency-domain loop behaviors effectively.

Feature importance plots generated by the pipeline confirmed these trends.

Across the 100 training loops:

- **Oscillation:**
    - 24 strong (“Yes”) cases
    - 18 mild/ambiguous (“Maybe”)
    - Total potential oscillatory loops: 42
- **Noise:**
    - 8 strong noisy loops

    - 19 mild ambiguous signals
    - Total: 27

- **Disturbance:**
    - Only 5 strong disturbance labels
      25 weak signals

This distribution shows that some behaviors—especially disturbance—are imbalanced. The use of RandomForest, XGBoost, and GradientBoosting mitigated this through robustness to imbalance.

The pipeline trained 3 models per classification task:

- **RandomForestClassifier**
- **XGBClassifier**
- **GradientBoostingClassifier**

And two models for regression:

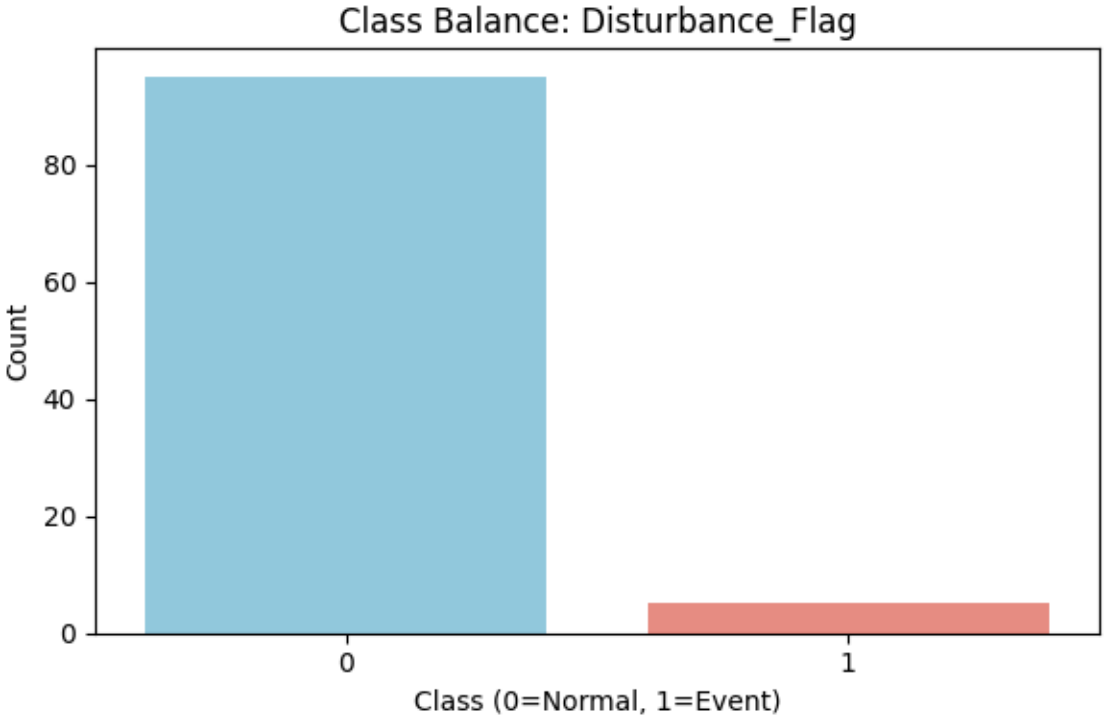
- **XGBRegressor**
- **SVR**

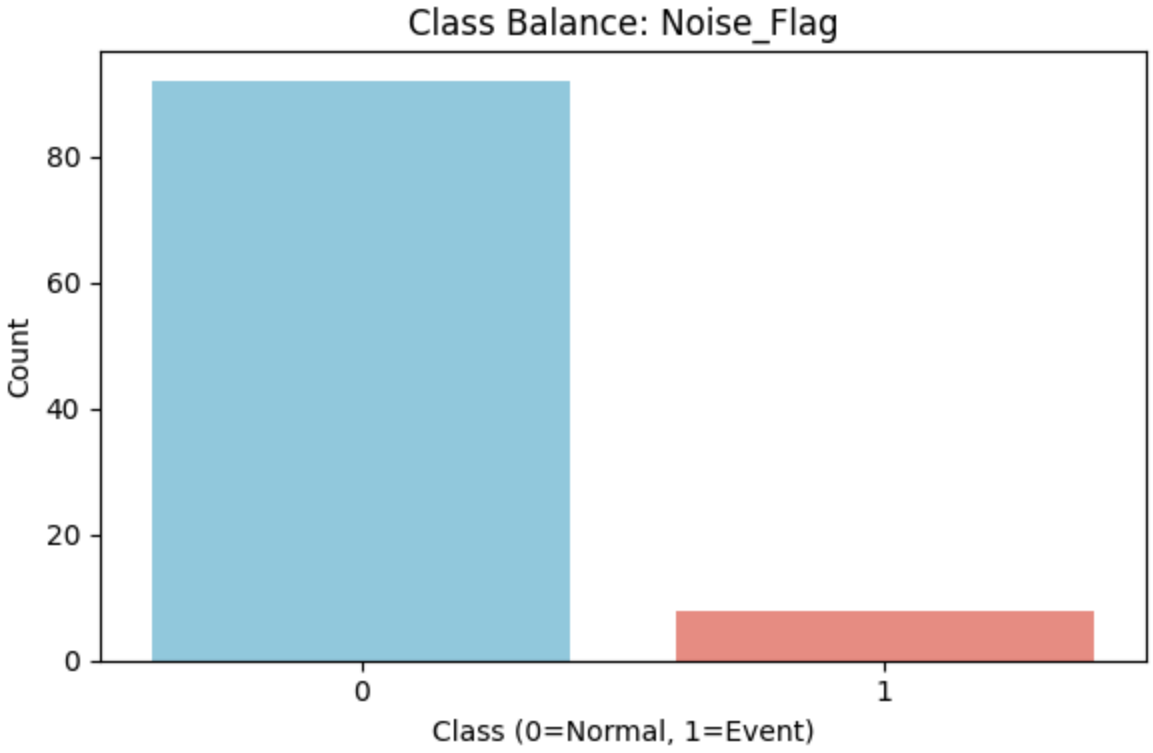
Using:

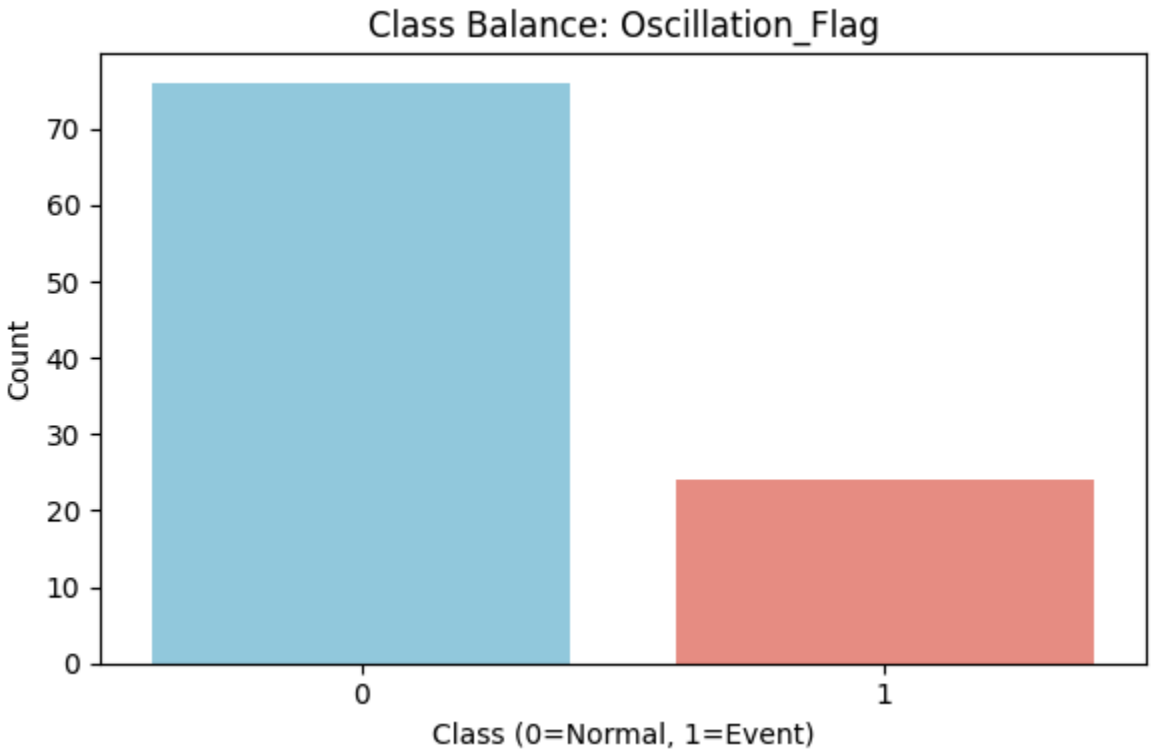
- 80:20 Train–Validation Split

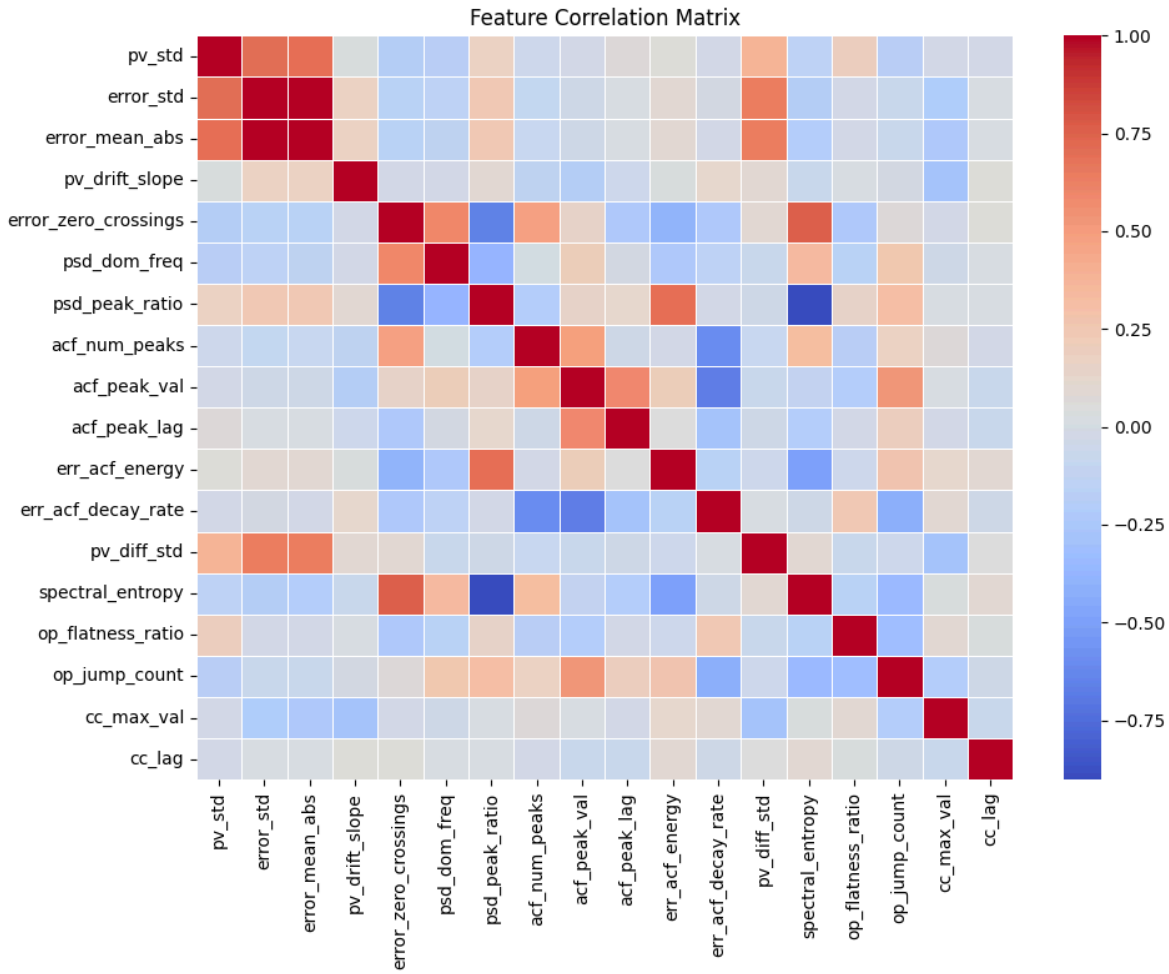
- 5-fold Cross-Validation

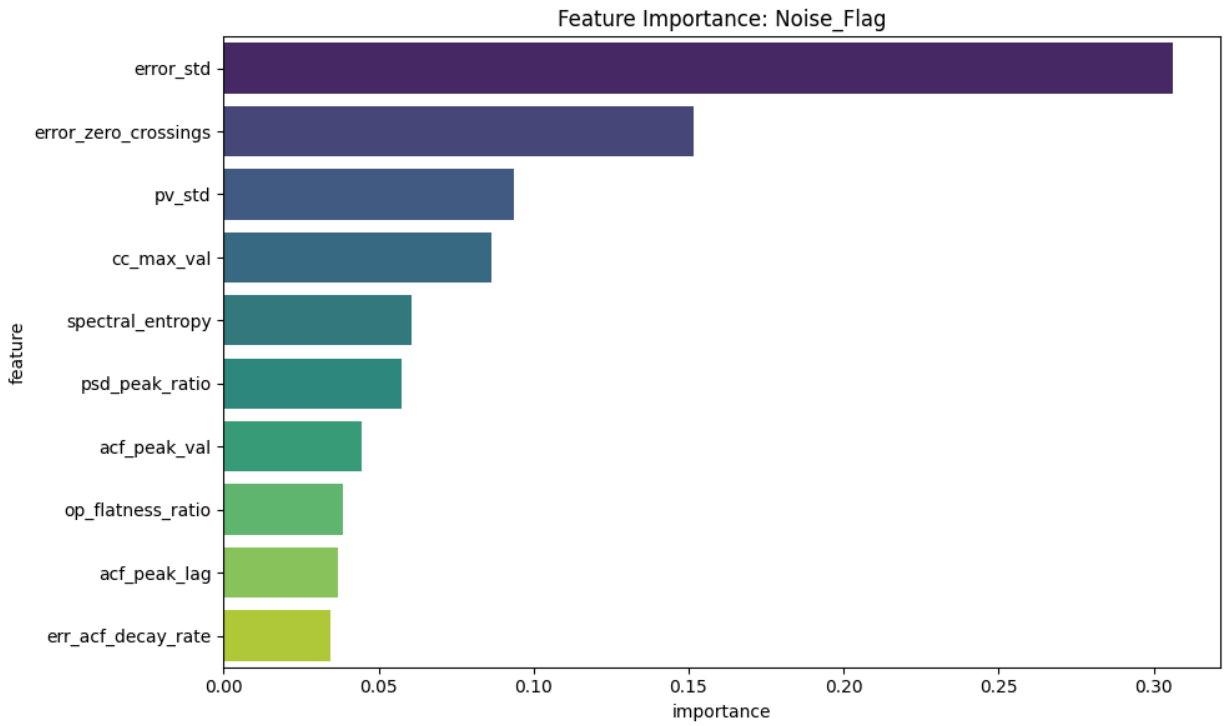
For each task, the pipeline automatically selected the best model based on F1-macro (classification) or RMSE (regression).

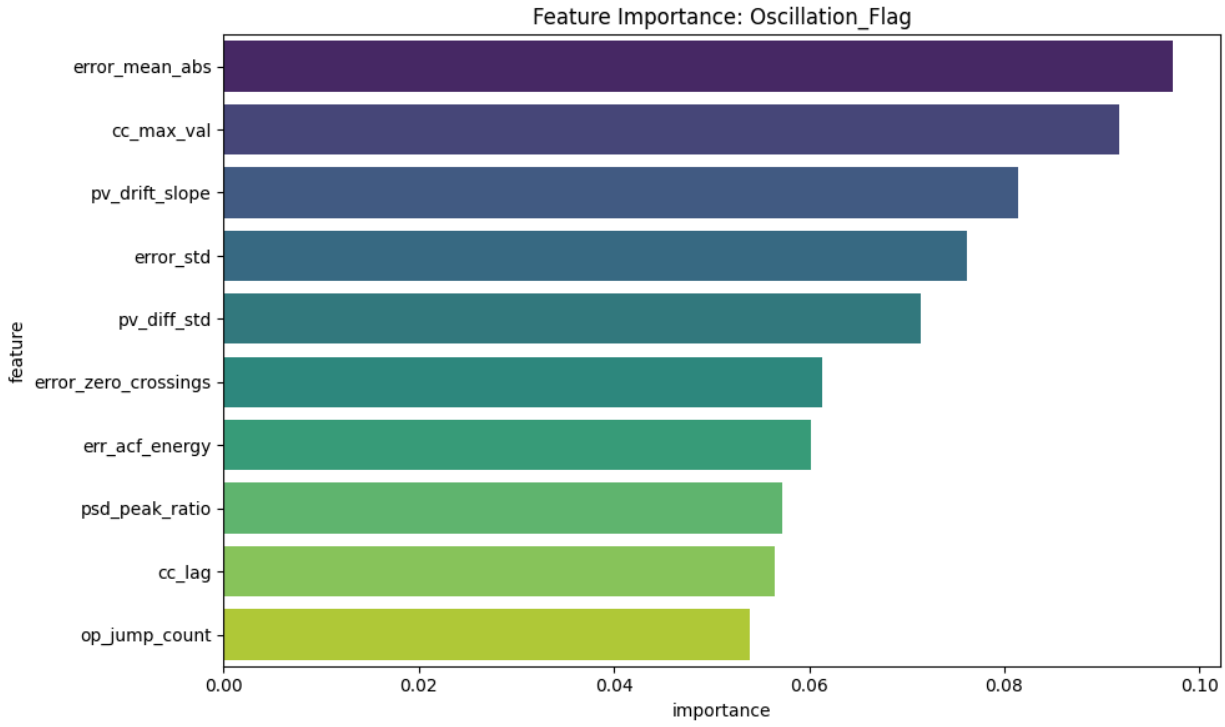

Class Balance: Disturbance_Flag
Count
0
20
40
60
80
0
1
Class (0=Normal, 1=Event)

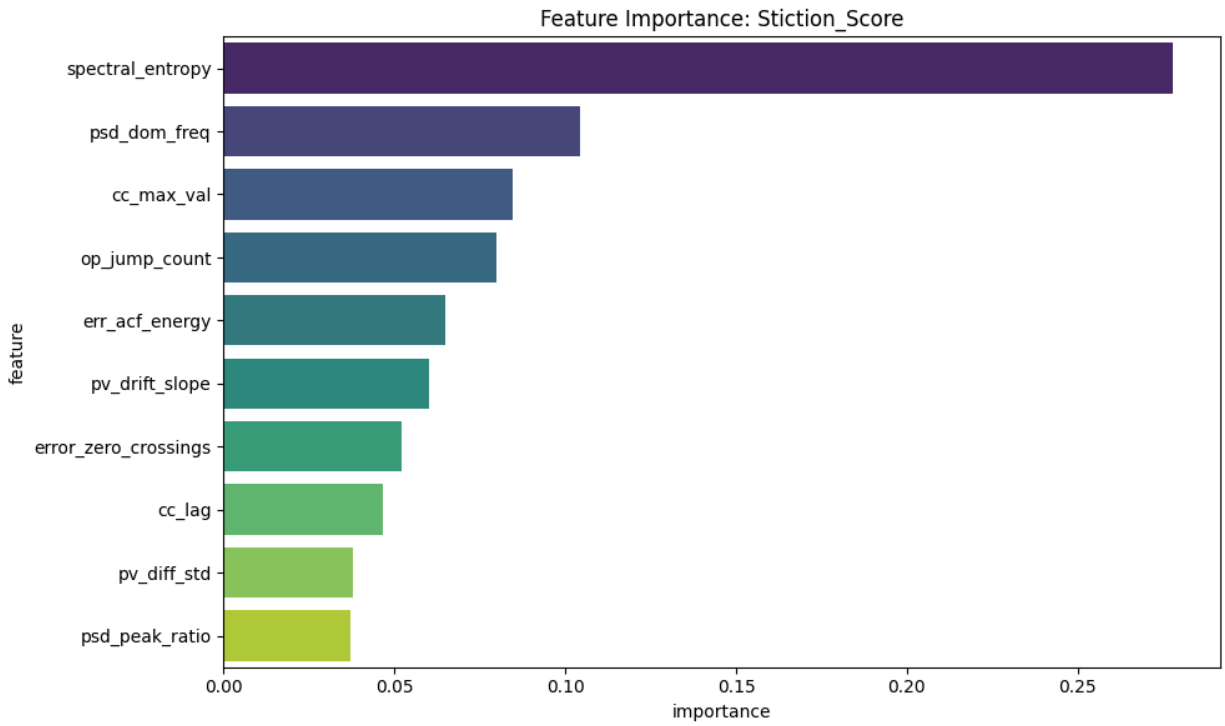

Class Balance: Noise_Flag
Count
0
20
40
60
80
0
1
Class (0=Normal, 1=Event)


Class Balance: Oscillation_Flag
Count
0
10
20
30
40
50
60
70
0
1
Class (0=Normal, 1=Event)


Feature Correlation Matrix
pv_std
error_std
error_mean_abs
pv_drift_slope
error_zero_crossings
psd_dom_freq
psd_peak_ratio
acf_num_peaks
acf_peak_val
acf_peak_lag
err_acf_energy
err_acf_decay_rate
pv_diff_std
spectral_entropy
op_flatness_ratio
op_jump_count
cc_max_val
cc_lag
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Feature Importance: Noise_Flag
error_std
error_zero_crossings
pv_std
cc_max_val
spectral_entropy
psd_peak_ratio
acf_peak_val
op_flatness_ratio
acf_peak_lag
err_acf_decay_rate
feature
0.00
0.05
0.10
0.15
0.20
0.25
0.30
importance


Feature Importance: Oscillation_Flag
error_mean_abs
cc_max_val
pv_drift_slope
error_std
pv_diff_std
error_zero_crossings
err_acf_energy
psd_peak_ratio
cc_lag
op_jump_count
feature
0.00
0.02
0.04
0.06
0.08
0.10
importance



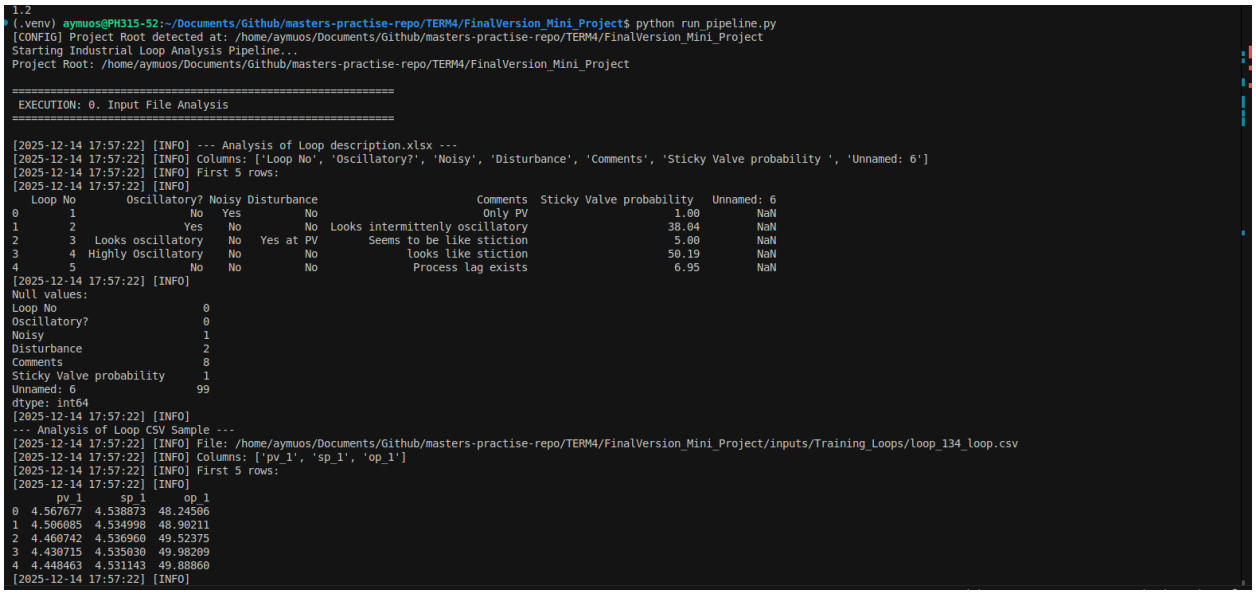

## RESULTS FROM UPLOADED LOOPS

| Loop_ID | oscillations | noise | disturbances | variance | probability_of_stiction | primary_inference | secondary_inference |
|---|---|---|---|---|---|---|---|
| 12 | 0.2708726734110887 | 0.1601600166741442 | 0.0197194249400882 | 1.5131496098282626 | 0.240135 | Low oscillation behavior detected. | Low noise levels detected. (Evidence: Entropy=1.85) Negligible disturbance detected. Low stiction levels detected. |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | (Evidence: Flatness=0.00, Jumps=0.061) |
| 101 | 0.44456107282589 5 | 0.353809156504668 | 0.00106028026607587 | 0.04312386799421 81 | 0.694956 | High probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.057) | Mild oscillation characteristics are also observed. (Evidence: ACF=0.55, ZC=0.16) Mild noise characteristics are also observed. (Evidence: Entropy=2.91) Negligible disturbance detected. |
| 131 | 0.31416266907551 8 | 0.014850170348840 7 | 0.00062898495732 7892 | 0.02565833571984 16 | 0.837710999999999 | High probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.069) | Mild oscillation characteristics are also observed. (Evidence: ACF=0.00, ZC=0.02) Negligible noise detected. (Evidence: Entropy=1.14) Negligible disturbance detected. |
| 77 | 0.22788124234909 3 | 0.158441401216801 | 0.00146417048275 475 | 0.055111472021091 5 | 0.32862 | Medium probability of valve | Low oscillation levels |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | stiction detected. (Evidence: Flatness=0.00, Jumps=0.063) | detected. (Evidence: ACF=0.00, ZC=0.04) Low noise levels detected. (Evidence: Entropy=1.89) Negligible disturbance detected. |
| | | | | | | | |

# 6. Computer Logs



```
1.2
(.venv) aymuos@PH315-52:~/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project$ python run_pipeline.py
[CONFIG] Project Root detected at: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project
Starting Industrial Loop Analysis Pipeline...
Project Root: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project

============================================================
 EXECUTION: 0. Input File Analysis
============================================================

[2025-12-14 17:57:22] [INFO] --- Analysis of Loop description.xlsx ---
[2025-12-14 17:57:22] [INFO] Columns: ['Loop No', 'Oscillatory?', 'Noisy', 'Disturbance', 'Comments', 'Sticky Valve probability ', 'Unnamed: 6']
[2025-12-14 17:57:22] [INFO] First 5 rows:
[2025-12-14 17:57:22] [INFO]
   Loop No        Oscillatory? Noisy Disturbance                      Comments  Sticky Valve probability   Unnamed: 6
0        1                  No   Yes          No                       Only PV                      1.00         NaN
1        2                 Yes    No          No  Looks intermittenly oscillatory                    38.04         NaN
2        3   Looks oscillatory    No   Yes at PV        Seems to be like stiction                     5.00         NaN
3        4  Highly Oscillatory    No          No             looks like stiction                    50.19         NaN
4        5                  No    No          No              Process lag exists                     6.95         NaN
[2025-12-14 17:57:22] [INFO]
Null values:
Loop No                      0
Oscillatory?                 0
Noisy                        1
Disturbance                  2
Comments                     8
Sticky Valve probability     1
Unnamed: 6                  99
dtype: int64
[2025-12-14 17:57:22] [INFO]
--- Analysis of Loop CSV Sample ---
[2025-12-14 17:57:22] [INFO] File: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/inputs/Training_Loops/loop_134_loop.csv
[2025-12-14 17:57:22] [INFO] Columns: ['pv_1', 'sp_1', 'op_1']
[2025-12-14 17:57:22] [INFO] First 5 rows:
[2025-12-14 17:57:22] [INFO]
      pv_1      sp_1      op_1
0  4.567677  4.538873  48.24506
1  4.506085  4.534998  48.90211
2  4.460742  4.536960  49.52375
3  4.430715  4.535030  49.98209
4  4.448463  4.531143  49.88860
[2025-12-14 17:57:22] [INFO]
```

```
(.venv) aymuos@PH315-52:~/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project$ python run pipeline.py
Statistics:
[2025-12-14 17:57:22] [INFO]
              pv_1         sp_1         op_1
count  5000.000000  5000.000000  5000.000000
mean      4.429446     4.429040    48.249760
std       0.159734     0.132140     1.387461
min       3.962856     4.110221    44.248040
25%       4.324970     4.317792    47.438365
50%       4.454554     4.486950    48.349230
75%       4.546883     4.536661    49.308903
max       4.729733     4.582000    51.148390

============================================================
 EXECUTION: 1. Training & Validation Pipeline
============================================================

Model: Found existing models at /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/models/trained_models.pkl
Skipping Training Step as requested. (Delete file to force retraining)

============================================================
 EXECUTION: 2. EDA Plot Generation (from Training Data)
============================================================

Generating EDA plots...
Loaded 100 rows.
Saved /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/eda/balance_Oscillation_Flag.png
Saved /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/eda/balance_Noise_Flag.png
Saved /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/eda/balance_Disturbance_Flag.png
Saved /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/eda/feature_correlation.png

============================================================
 EXECUTION: 3. Feature Importance Plots (from Models)
============================================================


[Feature Importance] Top 5 Features for Oscillation_Flag
       feature  importance
error_mean_abs    0.097280
    cc_max_val    0.091821
pv_drift_slope    0.081379
     error_std    0.076121
   pv_diff_std    0.071418
Plot saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/plots/eda/importance_Oscillation_Flag.png

[Feature Importance] Top 5 Features for Noise_Flag
             feature  importance
           error_std    0.306197
```

```
(.venv) aymuos@PH315-52:~/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project$ python run pipeline.py
[Feature Importance] Top 5 Features for Noise_Flag
             feature  importance
           error_std    0.306197
error_zero_crossings    0.151532
              pv_std    0.093692
          cc_max_val    0.086562
    spectral_entropy    0.060458
Plot saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/plots/eda/importance_Noise_Flag.png

[Feature Importance] Top 5 Features for Stiction_Score
         feature  importance
spectral_entropy    0.277817
    psd_dom_freq    0.104206
      cc_max_val    0.084547
   op_jump_count    0.079863
  err_acf_energy    0.064794
Plot saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/plots/eda/importance_Stiction_Score.png

============================================================
 EXECUTION: 4. Label Integrity & Count Analysis
============================================================

[2025-12-14 17:57:24] [INFO]
======================================
[2025-12-14 17:57:24] [INFO]  STEP: Label Integrity Audit
[2025-12-14 17:57:24] [INFO] ======================================
[2025-12-14 17:57:24] [INFO] Loaded Labels from: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/inputs/Loop description.xlsx
[2025-12-14 17:57:24] [INFO] Total Loops: 100

Audit: Oscillation
  - Strong Labels ('Yes'): 24
  - Weak Signals ('Maybe'): 18
  - Total Potential: 42

Audit: Noise
  - Strong Labels ('Yes'): 8
  - Weak Signals ('Maybe'): 19
  - Total Potential: 27

Audit: Disturbance
  - Strong Labels ('Yes'): 5
  - Weak Signals ('Maybe'): 25
  - Total Potential: 30

[Audit] Summary saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/label_audit_summary.csv

============================================================
```

```
(.venv) aymuos@PH315-52:~/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project$ python run_pipeline.py
[Audit] Summary saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/label_audit_summary.csv

============================================================
 EXECUTION: 5. Held-Out Test Data Inference
============================================================

=== Starting Test D                                    ining) ===
Loading models from Open file in editor (ctrl + click) 'Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/models/trained_models.pkl...
Models loaded from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/models/trained_models.pkl
Loading severity stats from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/severity_stats.json...
Found 4 CSV files in /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/test_data
Processing Loop: 12 (loop_12_loop.csv)...
  -> Inferences: Low oscillation behavior detected. ; Low noise levels detected. (Evidence: Entropy=1.85) Negligible disturbance detected.  Low stiction levels detected. (Evidence: Flatness=0.0
0, Jumps=0.061)
Processing Loop: 101 (loop_101_loop.csv)...
  -> Inferences: High probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.057) ; Mild oscillation characteristics are also observed. (Evidence: ACF=0.55, ZC=0.16) Mild noi
se characteristics are also observed. (Evidence: Entropy=2.91) Negligible disturbance detected.
Processing Loop: 131 (loop_131_loop.csv)...
  -> Inferences: High probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.069) ; Mild oscillation characteristics are also observed. (Evidence: ACF=0.00, ZC=0.02) Negligib
le noise detected. (Evidence: Entropy=1.14) Negligible disturbance detected.
Processing Loop: 77 (loop_77_loop.csv)...
  -> Inferences: Medium probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.063) ; Low oscillation levels detected. (Evidence: ACF=0.00, ZC=0.04) Low noise levels detected
. (Evidence: Entropy=1.89) Negligible disturbance detected.
Excel summary saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results.xlsx

Test Inference Complete. Results saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results.csv
Plots saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/test_data/debug_plots

============================================================
 EXECUTION: 6. Post-Process Categorization
============================================================

[2025-12-14 17:57:26] [INFO] === Post-Processing: Applying Logical Categories ===
[2025-12-14 17:57:26] [INFO] Loaded 4 rows from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results.csv
[2025-12-14 17:57:26] [INFO] Mapping categories...
[2025-12-14 17:57:26] [INFO]
Categorized results saved to:
[2025-12-14 17:57:26] [INFO]  - CSV: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results_categorized.csv
[2025-12-14 17:57:26] [INFO]  - Excel: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results_categorized.xlsx
[2025-12-14 17:57:26] [INFO]
[Preview]
[2025-12-14 17:57:26] [INFO]  Loop_ID  oscillations oscillation_category    noise noise_category  disturbances disturbance_category  variance  probability_of_stiction
                                    primary_inference
                                secondary_inference
      12      0.270873       No oscillation 0.160160       No noise      0.019719       No disturbance  1.513150                0.240135                                                Low
oscillation behavior detected.                                 Low noise levels detected. (Evidence: Entropy=1.85) Negligible disturbance detected.  Low stiction levels detected. (Evidence: Flat
ness=0.00, Jumps=0.061)
```


```
(.venv) aymuos@PH315-52:~/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project$ python run_pipeline.py
[Audit] Summary saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/label_audit_summary.csv

============================================================
 EXECUTION: 5. Held-Out Test Data Inference
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=== Starting Test D                                    ining) ===
Loading models from Open file in editor (ctrl + click) 'Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/models/trained_models.pkl...
Models loaded from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/models/trained_models.pkl
Loading severity stats from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/severity_stats.json...
Found 4 CSV files in /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/test_data
Processing Loop: 12 (loop_12_loop.csv)...
  -> Inferences: Low oscillation behavior detected. ; Low noise levels detected. (Evidence: Entropy=1.85) Negligible disturbance detected.  Low stiction levels detected. (Evidence: Flatness=0.0
0, Jumps=0.061)
Processing Loop: 101 (loop_101_loop.csv)...
  -> Inferences: High probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.057) ; Mild oscillation characteristics are also observed. (Evidence: ACF=0.55, ZC=0.16) Mild noi
se characteristics are also observed. (Evidence: Entropy=2.91) Negligible disturbance detected.
Processing Loop: 131 (loop_131_loop.csv)...
  -> Inferences: High probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.069) ; Mild oscillation characteristics are also observed. (Evidence: ACF=0.00, ZC=0.02) Negligib
le noise detected. (Evidence: Entropy=1.14) Negligible disturbance detected.
Processing Loop: 77 (loop_77_loop.csv)...
  -> Inferences: Medium probability of valve stiction detected. (Evidence: Flatness=0.00, Jumps=0.063) ; Low oscillation levels detected. (Evidence: ACF=0.00, ZC=0.04) Low noise levels detected
. (Evidence: Entropy=1.89) Negligible disturbance detected.
Excel summary saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results.xlsx

Test Inference Complete. Results saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results.csv
Plots saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/test_data/debug_plots

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 EXECUTION: 6. Post-Process Categorization
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[2025-12-14 17:57:26] [INFO] === Post-Processing: Applying Logical Categories ===
[2025-12-14 17:57:26] [INFO] Loaded 4 rows from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results.csv
[2025-12-14 17:57:26] [INFO] Mapping categories...
[2025-12-14 17:57:26] [INFO]
Categorized results saved to:
[2025-12-14 17:57:26] [INFO]  - CSV: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results_categorized.csv
[2025-12-14 17:57:26] [INFO]  - Excel: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/test_results_categorized.xlsx
[2025-12-14 17:57:26] [INFO]
[Preview]
[2025-12-14 17:57:26] [INFO]  Loop_ID  oscillations oscillation_category    noise noise_category  disturbances disturbance_category  variance  probability_of_stiction
                                    primary_inference
                                secondary_inference
      12      0.270873       No oscillation 0.160160       No noise      0.019719       No disturbance  1.513150                0.240135                                                Low
oscillation behavior detected.                                 Low noise levels detected. (Evidence: Entropy=1.85) Negligible disturbance detected.  Low stiction levels detected. (Evidence: Flat
ness=0.00, Jumps=0.061)
```

```
[Validation Summary: Raw Metrics vs Model Scores]
[2025-12-14 17:57:26] [INFO] Loop  RAW_ACF  RAW_Entropy RAW_Flatness  Model_Osc_Sev  Model_Stict_Prob Verdict
  12     0.45         1.67         2.2%       0.270873          0.240135   MATCH
 101     0.55         2.97         0.0%       0.444561          0.694956   MATCH
 131     0.81         1.14         0.2%       0.314163          0.837711   MATCH
  77     0.32         1.89         5.2%       0.227881          0.328620   MATCH
[2025-12-14 17:57:26] [INFO] Physical Validation Excel saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/physical_validation.xlsx

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 EXECUTION: 9. Training Data Reference Report
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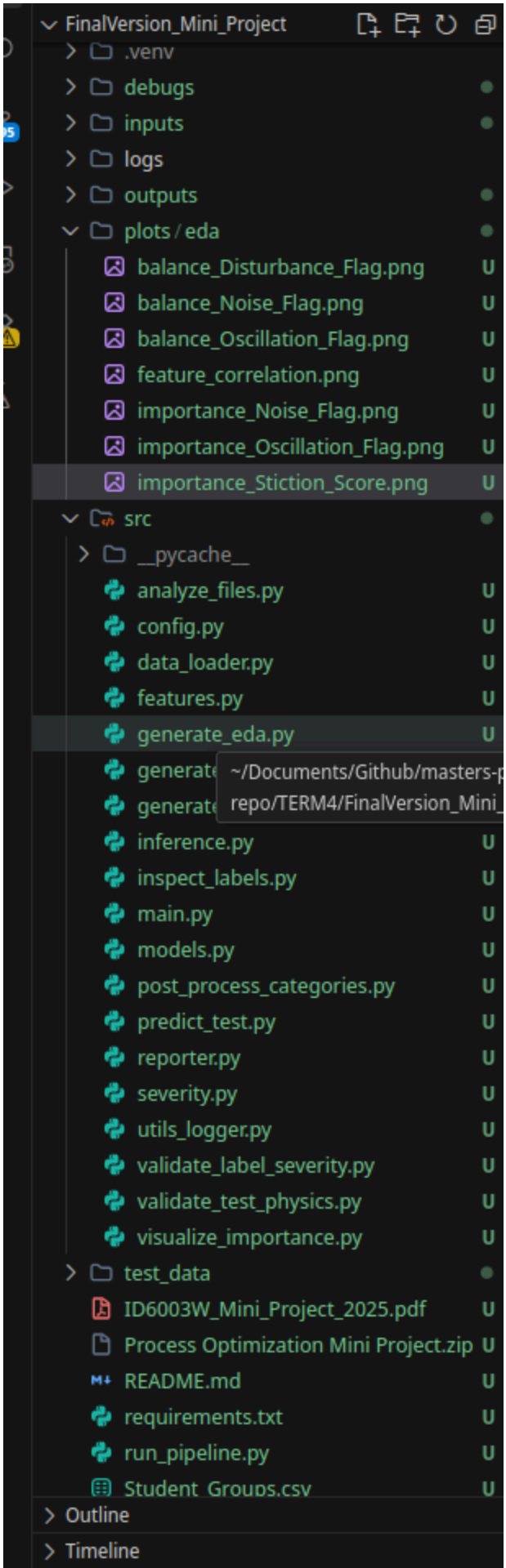
=== Generating Training Data Report (Excluding Test Loops) ===
Models loaded from /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/models/trained_models.pkl
Found 100 total files.
Excel report saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/training_data_summary.xlsx
Reference Training Data generated for 100 loops at /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/training_data_summary.csv

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 EXECUTION: 10. Severity vs Label Alignment
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[2025-12-14 17:57:31] [INFO] === Validating Severity vs Label Strength ===
[2025-12-14 17:57:31] [INFO] Processing 100 loops...
[2025-12-14 17:57:32] [INFO]
=== Validation Table: Severity Alignment ===
[2025-12-14 17:57:32] [INFO]    Category  Avg Severity (Strong=1)  Avg Severity (Weak=1)  Avg Severity (None)
Oscillation                0.297989               0.305857             0.403219
      Noise                0.512028               0.482459             0.246336
Disturbance                0.030096               0.159466             0.126415
[2025-12-14 17:57:32] [INFO]
Saved to /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs/severity_alignment.csv

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 EXECUTION: PIPELINE COMPLETE
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Total Execution Time: 9.95 seconds
Outputs available in: /home/aymuos/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project/outputs
(.venv) aymuos@PH315-52:~/Documents/Github/masters-practise-repo/TERM4/FinalVersion_Mini_Project$
```


FinalVersion_Mini_Project
.venv
debugs
inputs
logs
outputs
plots / eda
balance_Disturbance_Flag.png U
balance_Noise_Flag.png U
balance_Oscillation_Flag.png U
feature_correlation.png U
importance_Noise_Flag.png U
importance_Oscillation_Flag.png U
importance_Stiction_Score.png U
src
__pycache__
analyze_files.py U
config.py U
data_loader.py U
features.py U
generate_eda.py U
~/Documents/Github/masters-p
repo/TERM4/FinalVersion_Mini_
inference.py U
inspect_labels.py U
main.py U
models.py U
post_process_categories.py U
predict_test.py U
reporter.py U
severity.py U
utils_logger.py U
validate_label_severity.py U
validate_test_physics.py U
visualize_importance.py U
test_data
ID6003W_Mini_Project_2025.pdf U
Process Optimization Mini Project.zip U
README.md U
requirements.txt U
run_pipeline.py U
Student_Groups.csv U
Outline
Timeline

How to run :  install requirements.txt , and execute src/run_pipeline.py . Ensure test files are in test_data